

BINGLEY, United Kingdom

WMO: 033440

Lat: 53.8113N

Lon: 1.8668W

Elev: 267

StdP: 98.16

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.8	-2.5	-6.4	2.3	-1.9	-5.0	2.6	-1.3	14.9	6.5	13.6	6.6	2.9	290	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	24.0	17.4	21.9	16.4	20.1	15.5	18.3	22.0	17.2	20.6	16.2	19.1	3.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.4	19.8	15.9	11.6	18.4	15.0	11.0	17.3	52.8	22.2	49.2	20.6	46.4	19.2	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.3	9.0	DB	-5.8	27.3	2.0	2.6	-7.2	29.1	-8.4	30.6	-9.5	32.1	-11.0	34.0	(4)
(5)			WB	-6.3	20.6	1.9	2.0	-7.7	22.1	-8.8	23.2	-9.8	24.3	-11.2	25.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	3.5	3.6	4.8	7.2	10.1	13.0	15.1	14.7	12.4	9.7	6.0	4.0	(6)
(7)		DBStd	5.04	2.86	3.02	2.99	2.96	3.00	2.63	2.51	2.35	2.30	2.68	2.86	3.23	(7)
(8)		HDD10.0	1026	202	181	162	93	36	4	0	0	4	38	122	185	(8)
(9)		HDD18.3	3525	460	414	419	335	256	162	107	116	177	269	369	443	(9)
(10)		CDD10.0	554	0	1	1	9	38	93	157	147	77	28	3	1	(10)
(11)		CDD18.3	11	0	0	0	0	0	1	5	4	1	0	0	0	(11)
(12)		CDH23.3	86	0	0	0	0	2	16	41	26	2	1	0	0	(12)
(13)		CDH26.7	8	0	0	0	0	0	1	6	2	0	0	0	0	(13)
(14)	Wind	WSAvg	4.2	5.1	5.2	4.7	4.1	3.9	3.5	3.3	3.5	4.1	4.5	4.9	(14)	
(15)	Precipitation	PrecAvg	1039	113	85	65	65	67	73	78	86	79	98	108	123	(15)
(16)		PrecMax	1412	234	224	136	146	134	233	171	221	163	204	246	272	(16)
(17)		PrecMin	753	14	19	7	12	15	14	35	14	25	31	44	21	(17)
(18)		PrecStd	180	52	49	34	36	31	51	36	42	38	43	54	58	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	12.8	14.2	19.4	22.8	25.5	26.9	26.5	22.7	18.2	13.8	12.2	(19)
(20)			MCWB	10.0	9.8	9.8	12.9	16.7	17.8	18.8	17.6	17.7	14.9	12.8	11.1	(20)
(21)		2%	DB	9.9	10.4	11.9	15.8	20.1	22.1	24.2	22.8	19.5	15.9	12.4	11.1	(21)
(22)			MCWB	8.9	8.9	9.3	11.4	14.4	16.0	17.6	17.4	15.2	13.7	11.3	10.2	(22)
(23)		5%	DB	8.7	9.1	10.5	13.7	17.4	19.9	22.1	20.4	17.6	14.7	11.2	9.8	(23)
(24)			MCWB	7.8	7.9	8.5	10.3	12.9	14.7	16.6	16.2	14.2	13.1	10.2	8.9	(24)
(25)		10%	DB	7.8	8.0	9.2	12.0	15.2	18.0	20.0	18.8	16.2	13.7	10.2	8.7	(25)
(26)			MCWB	6.9	6.8	7.6	9.3	11.6	13.7	15.7	15.3	13.5	12.3	9.3	7.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	10.4	10.9	13.8	17.5	18.7	20.7	19.5	18.1	15.7	12.9	11.3	(27)
(28)			MCD B	10.9	11.8	12.5	17.3	21.6	23.5	24.8	22.8	21.4	16.9	13.7	11.9	(28)
(29)		2%	WB	9.0	9.1	9.7	12.1	15.0	16.7	18.5	17.8	16.3	14.4	11.6	10.3	(29)
(30)			MCD B	9.7	10.2	11.4	15.0	19.0	20.8	22.4	21.0	18.5	15.5	12.2	11.0	(30)
(31)		5%	WB	8.0	8.0	8.8	10.8	13.5	15.4	17.2	16.8	15.0	13.4	10.4	9.1	(31)
(32)			MCD B	8.7	9.0	10.4	13.4	16.4	18.7	20.6	19.8	16.4	14.5	11.0	9.7	(32)
(33)		10%	WB	6.9	6.9	7.8	9.5	12.2	14.4	16.1	16.0	14.0	12.4	9.4	7.9	(33)
(34)			MCD B	7.7	7.9	9.1	11.6	14.7	17.3	19.1	18.2	15.6	13.4	10.0	8.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.0	4.6	5.7	7.1	7.6	7.5	7.6	7.0	6.5	5.4	4.3	4.2	(35)
(36)			MCDBR	5.3	5.7	7.4	9.7	11.3	11.2	9.7	8.7	6.3	4.9	5.4	(36)	
(37)			MCWBR	5.3	4.9	5.4	6.0	6.9	6.0	5.8	5.1	5.2	4.6	4.7	5.1	(37)
(38)		5% WB	MCD BR	5.2	5.5	6.8	9.0	10.1	9.9	9.7	8.6	7.4	5.9	4.9	5.2	(38)
(39)			MCWBR	5.3	5.0	5.4	6.1	6.7	6.0	5.9	5.1	5.4	4.9	4.9	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.328	0.357	0.382	0.383	0.384	0.384	0.371	0.354	0.338	0.313	0.303	(40)	
(41)		taud	2.322	2.382	2.312	2.262	2.297	2.317	2.317	2.369	2.426	2.444	2.390	2.304	(41)	
(42)		Ebn at Noon	643	768	819	844	863	864	856	847	817	748	645	574	(42)	
(43)		Edh at Noon	59	76	100	119	121	120	118	106	89	70	54	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.63	1.30	2.40	3.75	4.76	4.91	4.73	3.89	2.81	1.57	0.78	0.49	(44)	
(45)		RadStd	0.07	0.15	0.26	0.40	0.32	0.45	0.47	0.26	0.24	0.14	0.08	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page